Any stars that happened to be formed within the Milky Way cloud before the collapse maintained large, plunging orbits. The remaining gas, which easily sticks to itself, like a mid-air collision of two hot marshmallows, got pinned at the mid-plane and is responsible for all subsequent generations of stars, including the Sun. The current Milky Way, which is neither collapsing nor expanding, is a gravitationally mature system where one can think of the orbiting stars above and below the disk as the skeletal remains of the original spherical gas cloud.

This general flattening of objects that rotate is why Earth's pole-to-pole diameter is smaller than its diameter at the equator. Not by much: three-tenths of one percent—about twenty-six miles. But Earth is small, mostly solid, and doesn't rotate all that fast. At twenty-four hours per day, Earth carries anything on its equator at a mere 1,000 miles per hour. Consider the jumbo, fast-rotating, gaseous planet Saturn. Completing a day in just ten and a half hours, its equator revolves at 22,000 miles per hour and its pole-to-pole dimension is a full ten percent flatter than its middle, a difference noticeable even through a small amateur telescope. Flattened spheres are more generally called oblate spheroids, while spheres that are elongated pole-to-pole are called prolate. In everyday life, hamburgers and hot dogs make excellent (although somewhat extreme) examples of each shape. I don't know about you, but the planet Saturn pops into my mind with every bite of a hamburger I take.

✴

We use the effect of centrifugal forces on matter to offer insight into the rotation rate of extreme cosmic objects. Consider pulsars. With some rotating at upward of a thousand revolutions per second, we know that they cannot be made of household ingredients, or they would spin themselves apart. In fact, if a pulsar rotated any faster, say 4,500 revolutions per second, its equator would be moving at the speed of light, which tells you that this material is unlike any other. To picture a pulsar, imagine the mass of the Sun packed into a ball the size of Manhattan. If that's hard to do, then maybe it's easier if you imagine stuffing about a hundred million elephants into a Chapstick casing. To reach this density, you must compress all the empty space that atoms enjoy around their nucleus and among their orbiting electrons. Doing so will crush nearly all (negatively charged) electrons into (positively charged) protons, creating a ball of (neutrally charged) neutrons with a crazy-high surface gravity. Under such conditions, a neutron star's mountain range needn't